

TSA Meeting — Slide 5 Study Note

Ch 2: Unified Inverse-Estimation Framework

Mathematical Foundations · LM Update · Confidence Gate · Theorem 2.1

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Slide Overview and Narrative

What This Slide Shows

Slide 5 presents the **mathematical spine** of the entire thesis. Every adaptive algorithm in Chapters 4–9 (87L, SyncOC, 87T, Distance, Microgrid, Digital Twin) is an instantiation of the same three-layer structure:

1. **Inverse estimator:** The LM update rule fits a physical model $\mathbf{x}(t; \boldsymbol{\theta})$ to real-time phasor measurements, identifying latent protection parameters $\boldsymbol{\theta} = \{k_{\text{ibr}}, Z_{\text{line}}, Y_{\text{ibr}}\}$ that cannot be measured directly by any relay.
2. **Confidence gate (Stage-2):** A scalar score $\gamma = \exp(-\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*\|^2 / 2\sigma_{\hat{\boldsymbol{\theta}}}^2)$ decides whether the estimate is trustworthy enough to update relay settings. Only estimates with $\gamma \geq \gamma_{\min}$ pass through.
3. **Bounded update law:** A step-limited, clipped projection ensures that even a wrong estimate that passes the gate cannot push settings outside the safe range defined at commissioning.

The **Mathematical Foundations table** on the right maps each sub-problem to its solver: LM for parameter update, EKF for state estimation, Wald SPRT for fault detection, Physics-Informed NN for ML classification, and projected gradient for constrained optimisation.

Theorem 2.1 provides the formal guarantee: under bounded IBR admittance and $\lambda > 0$, the LM update converges to a neighbourhood of the true parameters with radius $\mathcal{O}(\sigma_{\text{noise}})$. This is the thesis's mathematical bedrock — every claim about algorithm performance in later chapters depends on it.

Slide at a Glance: The Five Components

Item on Slide	Mathematical Object	Purpose
LM update equation	$\hat{\boldsymbol{\theta}}_{t+1} = \hat{\boldsymbol{\theta}}_t - \alpha_t J^\top (J J^\top + \lambda I)^{-1} \mathbf{r}(\hat{\boldsymbol{\theta}}_t)$	Solves nonlinear least-squares online
Parameter vector $\boldsymbol{\theta}$	$\{k_{\text{ibr}}, Z_{\text{line}}, Y_{\text{ibr}}\}$	Latent (unobservable) relay parameters
Residual $\mathbf{r}$	$\mathbf{y} - \mathbf{x}(\boldsymbol{\theta})$	Fit error between model and phasor data
Bounded update α_t	$\alpha_t \propto (1 - k_{\text{ibr}}^2)$	Freezes adaptation at high IBR penetration
Confidence gate γ	$\exp(-\ \hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*\ ^2 / 2\sigma_{\hat{\boldsymbol{\theta}}}^2) \geq \gamma_{\min}$	Authorises relay setting update

Engineering Motivation: Why a Unified Framework?

The Unification Argument

Without a unified mathematical framework, each protection element would need:

- Its own estimation algorithm (risk of inconsistent performance guarantees).
- Its own confidence metric (no common threshold for “trustworthy”).
- Its own safety mechanism (risk of unsafe settings in one element while another is conservative).

By casting all problems as *instances of the same inverse estimation problem* (Eq. (??)), the thesis achieves:

- **One proof:** Theorem 2.1 covers all algorithms.
- **One threshold:** $\gamma_{\text{th}} = 0.70$ (SyncOC) or 0.85 (87L) is the only design parameter per element.
- **One update mechanism:** bounded clipped projection, identical in structure across all relay types.

The only things that change per chapter are: the physical model $\mathbf{x}(t; \boldsymbol{\theta})$ and the parameter vector $\boldsymbol{\theta}$. Everything else — LM, confidence gate, bounded update — is inherited from Chapter 2.

Technical Deep-Dive: The LM Update Rule

Standard vs. Slide Form

The slide shows the LM step as:

$$\hat{\boldsymbol{\theta}}_{t+1} = \hat{\boldsymbol{\theta}}_t - \alpha_t J^\top (J J^\top + \lambda I)^{-1} \mathbf{r}(\hat{\boldsymbol{\theta}}_t) \quad (1)$$

The standard form (used in the chapter’s Algorithm 2.1) is:

$$\boldsymbol{\delta} = -(J^\top J + \mu I)^{-1} J^\top \mathbf{r} \quad (2)$$

These are **algebraically identical** by the Woodbury matrix identity:

$$\underbrace{J^\top (J J^\top + \lambda I)^{-1}}_{\text{slide form: } (n_K) \times p} = \underbrace{(J^\top J + \lambda I)^{-1} J^\top}_{\text{standard form: } p \times (n_K)} \quad (3)$$

The slide’s form (left-multiply by $J^\top$) solves a system of size $n_K \times n_K$ (number of measurements), while the standard form solves a $p \times p$ system (number of parameters). When $p \ll n_K$ (6 parameters, hundreds of samples), the standard form is cheaper. When $n_K \ll p$ (the inverse problem is underdetermined), the slide’s form is preferred.

Condition Number Reduction: Why LM Damping λ Matters

The condition number of the undamped normal equations is:

$$\kappa(J^\top J) = \kappa(J)^2$$

LM damping adds λI , yielding:

$$\kappa(J^\top J + \lambda I) = \frac{\sigma_{\max}^2(J) + \lambda}{\sigma_{\min}^2(J) + \lambda} \leq \kappa(J)^2$$

As $\lambda \rightarrow \infty$: $\kappa \rightarrow 1$ (perfectly conditioned gradient descent).

As $\lambda \rightarrow 0$: $\kappa \rightarrow \kappa(J)^2$ (Gauss-Newton, fastest near solution).

The chapter's adaptive μ schedule (Section 3.3) interpolates: large μ far from minimum, small μ near minimum. This is why LM achieves superlinear convergence (between gradient descent's linear and Newton's quadratic).

Residual $\mathbf{r}(\hat{\boldsymbol{\theta}})$: The Physical Model Connection

The residual is:

$$\mathbf{r}(\boldsymbol{\theta}) = \mathbf{y}_{\text{measured}} - \mathbf{x}(t; \boldsymbol{\theta}) \quad (4)$$

In each chapter, $\mathbf{x}(t; \boldsymbol{\theta})$ is a specific physical model:

Chapter	Model $\mathbf{x}(t; \boldsymbol{\theta})$	Parameters $\boldsymbol{\theta}$
4 (87L)	IBR differential current model $I_{\text{diff}}(t; k_{\text{ibr}}, \phi_B)$	k_{ibr}, ϕ_B
5 (SyncOC)	6-param fault current: $I_{ss} + I_{sub}e^{-t/\tau_{ac}} \cos(\omega_0 t + \phi_a)$	$\{I_{ss}, I_{sub}, \tau_{ac}, \tau_{dc}, \phi_a, t_0\}$
9 (DT)	IBR Thevenin admittance $Y_{\text{ibr}}(\omega)$	$k_{\text{ibr}}, Z_{\text{line}}, Y_{\text{ibr}}$
6 (87T)	Inrush/fault harmonic envelope	$\{i_2/i_1, \phi_{\text{hm}}\}$

In the digital twin (Ch 9), $\boldsymbol{\theta} = \{k_{\text{ibr}}, Z_{\text{line}}, Y_{\text{ibr}}\}$ which is precisely what the slide labels as the “latent protection parameters.”

Jacobian J : Sensitivity Analysis

The Jacobian $J_{ij} = \partial x_i / \partial \theta_j$ measures how sensitive each measurement x_i is to each parameter θ_j . In the protection context:

- $\partial I_{\text{diff}} / \partial k_{\text{ibr}}$: How the differential current changes with IBR penetration — large early in the fault window.
- $\partial i_a(t) / \partial \tau_{ac}$: How the waveform tail changes with the AC time constant — large at $t \gg \tau_{ac}$, identifiable only in the tail of the measurement window.

The two-pass LM strategy (Pass 1 for fast params, Pass 2 for slow params) exploits this structure to reduce the effective Jacobian condition number $\kappa_n(J)$, which directly improves the confidence score component γ_J .

Bounded Update Law: $\alpha_t \propto (1 - k_{\text{ibr}}^2)$

Why the Step Size Depends on k_{ibr}

The slide states: “Bounded update law: $\alpha_t \propto (1 - k_{\text{ibr}}^2)$ prevents parameter drift in healthy state.”

Interpretation:

- $k_{\text{ibr}} = 0$ (no IBR): $\alpha_t \propto 1$ — maximum step size. The system is purely SG-dominated and the fault current model is most identifiable; allow fast adaptation.
- $k_{\text{ibr}} = 0.5$: $\alpha_t \propto 0.75$ — moderate adaptation.
- $k_{\text{ibr}} \rightarrow 1.0$ (pure IBR): $\alpha_t \rightarrow 0$ — adaptation freezes. At full IBR saturation, the fault current is at the controlled limit and carries no new information about the network state. Continuing to update θ would cause *parameter drift* — random walk due to measurement noise.

Protection security implication: When $k_{\text{ibr}} \approx 1$ (the most challenging operating point for protection), the relay settings are *not changed*. This is the conservative choice: keep the last valid settings rather than risk a bad update under the most uncertain conditions.

The general bounded update (chapter Eq. 2.16) at each relay setting ξ_k :

$$\xi_k^{(n+1)} = \text{clip} \left(\xi_k^{(n)} + \delta_{\max} \text{sgn}(\hat{\xi}_k - \xi_k^{(n)}) \cdot \min \left(1, \frac{|\hat{\xi}_k - \xi_k^{(n)}|}{\delta_{\max}} \right), \xi_{\min}, \xi_{\max} \right) \quad (5)$$

This guarantees (Proposition 2.1): if $\xi_k^{(0)} \in [\xi_{\min}, \xi_{\max}]$, then $\xi_k^{(n)} \in [\xi_{\min}, \xi_{\max}]$ for all $n \geq 0$. Combined with $\alpha_t \propto (1 - k_{\text{ibr}}^2)$, this means:

$$\|\hat{\theta}^{(n+1)} - \hat{\theta}^{(n)}\| \leq \alpha_t \cdot C_{\text{LM}}(1 - k_{\text{ibr}}^2) \quad (6)$$

The step is bounded by both the LM convergence radius C_{LM} and the IBR-dependent gain $(1 - k_{\text{ibr}}^2)$.

Stage-2 Confidence Gate: Two Formulations

Theoretical Form (Slide)

The slide shows the Gaussian confidence score:

$$\gamma = \exp \left(-\frac{\|\hat{\theta} - \theta^*\|^2}{2\sigma_{\theta}^2} \right) \geq \gamma_{\min} \quad (7)$$

This is the **theoretical motivation**: if the estimation error $\hat{\theta} - \theta^*$ follows a zero-mean Gaussian with covariance $\sigma_{\theta}^2 I$ (i.e., estimation noise is Gaussian), then γ is the likelihood of the current estimate being correct. $\gamma = 1$ at perfect estimate; $\gamma \rightarrow 0$ as the error grows.

Practical Implementation (Chapter)

The chapter's four-component score (Eq. 2.8) is the **operational approximation**:

$$\gamma = w_1\gamma_R + w_2\gamma_J + w_3\gamma_B + w_4\gamma_P \quad (8)$$

with weights $[w_1, w_2, w_3, w_4] = [0.35, 0.30, 0.20, 0.15]$:

Component	Formula	Range	What It Checks
γ_R	$1/(1 + 100(1 - R^2))$	$[0,1]$	Normalised fit quality ($R^2 \rightarrow 1$)
γ_J	$1/(1 + 0.1 \log_{10} \kappa_n)$	$[0,1]$	Jacobian condition number (identifiability)
γ_B	$1 - N_{\text{bounds}}/p$	$[0,1]$	Fraction of params at physical bounds
γ_P	$1 - N_{\text{implausible}}/p$	$\{0,0.5,1\}$	Physics plausibility check

Connection to the Gaussian form: The component γ_R approximates $\exp(-(1 - R^2)/\sigma_R^2)$ for small $(1 - R^2)$ — i.e., the Gaussian form evaluated at the residual magnitude. Components γ_J , γ_B , γ_P are engineering guards that do not have direct Gaussian analogues but are necessary for protection-grade reliability.

Threshold Values Across SAMBP Elements

Protection Element	γ_{th}	Reason for Level
SAMBP-SyncOC (87OC)	0.70	Generator OC is backup; looser gate acceptable
SAMBP-87L (line diff)	0.85	Primary protection; must be conservative
SAMBP-87T (transformer)	0.80	Primary; 87T false trip is a transformer outage
Digital Twin EKF	0.75	State tracking; moderate conservatism

Theorem 2.1 — Convergence Guarantee

Theorem 2.1 (Convergence) — Formal Statement

Conditions:

- (i) IBR admittance is bounded: $|Y_{\text{ibr}}| \leq Y_{\text{max}} < \infty$.
- (ii) LM damping: $\lambda > 0$ (strictly positive).
- (iii) The model $\mathbf{x}(t; \boldsymbol{\theta})$ is Lipschitz in $\boldsymbol{\theta}$ with constant L_θ .
- (iv) Measurement noise: $\mathbf{v}_k \sim \mathcal{N}(\mathbf{0}, \sigma_{\text{noise}}^2 I)$, i.i.d. and independent of $\boldsymbol{\theta}$.

Conclusion: The LM update sequence $\{\hat{\boldsymbol{\theta}}^{(n)}\}$ satisfies:

$$\limsup_{n \rightarrow \infty} \mathbb{E} \left[\|\hat{\boldsymbol{\theta}}^{(n)} - \boldsymbol{\theta}^*\|^2 \right] = \mathcal{O}(\sigma_{\text{noise}}^2) \quad (9)$$

i.e., the expected squared estimation error is asymptotically bounded by a constant times the noise variance. The algorithm converges to a *ball* around the true parameters, not necessarily to the exact true value (which is unattainable with noisy measurements).

What Theorem 2.1 Does NOT Guarantee

- **Not global convergence:** LM may converge to a local minimum if $\mathbf{x}(t; \boldsymbol{\theta})$ is highly non-convex. The two-pass strategy and physical bounds mitigate this.
- **Not finite-sample:** The bound is asymptotic ($n \rightarrow \infty$). For the protection application, “convergence” means within the fault window (≤ 10 LM iterations ≈ 3 ms). The empirical convergence to $R^2 > 0.999$ in 15–20 iterations is stronger than what Theorem 2.1 requires.
- **Not on the confidence gate:** Theorem 2.1 bounds the estimator; the gate $\gamma \geq \gamma_{\text{th}}$ is a separate mechanism that further reduces the probability of acting on a bad estimate.

Mathematical Foundations Table: Five Methods

The Five Methods and Why Each Was Chosen

Levenberg–Marquardt (Parameter update): LM is the gold standard for bounded nonlinear least-squares. Unlike pure gradient descent, it uses curvature information (Jacobian) for faster convergence. Unlike pure Newton’s method, the λI regularisation prevents divergence on ill-conditioned problems (which arise at low k_{ibr} where fault current is weak).

Extended Kalman Filter (State estimation): The EKF is the recursive counterpart to LM. It tracks time-varying parameters ($k_{\text{ibr}}(t)$, $Z_{\text{line}}(t)$) continuously as they evolve between faults. The EKF prediction step provides implicit Tikhonov regularisation equivalent to $\lambda = Q^{-1}$ (process noise covariance), connecting formally to the LM framework.

Wald SPRT (Fault detection): The sequential probability ratio test is the *optimal* fixed-sample test for a binary hypothesis (fault vs. no fault) in the Neyman–Pearson sense. It minimises expected sample number subject to specified false-alarm and miss-detection probabilities. Used in the 87T element to replace the fixed 15% second-harmonic threshold.

Physics-Informed NN (ML classification): The PINN adds physics constraints (KVL/KCL) to the neural network loss function, preventing the network from learning physically impossible patterns. Used in the ML classification module (Ch 10) for fault-type identification. Without physics constraints, networks trained on simulation data fail on corner-case real measurements.

Projected gradient (Optimisation): The projected-gradient descent enforces the CTI coordination constraint $t_{\text{primary}} + \text{CTI} \leq t_{\text{backup}}$ during each relay setting update. Every gradient step is projected back onto the feasible set $\mathcal{C}$ defined by the coordination inequality. This is the mechanism by which SAMBP-SyncOC maintains relay coordination automatically as k_{ibr} changes.

Cross-Thesis Connections

Chapter	Which Ch 2 Component It Uses	Ch 2 Equation Used
Ch 4 (87L)	LM estimator identifies k_{ibr} from I_{diff} waveform; confidence gate threshold $\gamma_{th} = 0.85$; bounded update of bias slope k_m^*	Eq. (1); $\gamma_{th} = 0.85$
Ch 5 (SyncOC)	Two-pass LM for 6-parameter SG fault current; $\gamma_{th} = 0.70$; projected-gradient CTI constraint	Eq. (2); Eq. (5)
Ch 7 (Distance)	LM identifies k_{ibr} from pre-fault admittance; updates Zone 1 correction factor k_X each 10 ms	Eq. (1); $\alpha_t \propto (1 - k_{ibr}^2)$
Ch 9 (Digital Twin)	EKF (recursive LM) tracks $\theta = \{k_{ibr}, Z_{line}, Y_{ibr}\}$ continuously between faults	Formal equivalence: EKF $\leftrightarrow$ LM + process noise
Ch 10 (PINN)	Physics-informed loss adds KVL/KCL penalty to NN training	PINN constraint from “Mathematical Foundations” table
Ch 6 (87T)	Wald SPRT replaces fixed harmonic threshold; LM identifies harmonic envelope parameters	SPRT + LM for inrush parameter identification

Key Equations Reference Card

All Equations You Must Know for This Slide

LM update (slide form):
$$\hat{\boldsymbol{\theta}}_{t+1} = \hat{\boldsymbol{\theta}}_t - \alpha_t J^\top (J J^\top + \lambda I)^{-1} \mathbf{r}(\hat{\boldsymbol{\theta}}_t)$$
 (E1)

LM step (standard form):
$$\boldsymbol{\delta} = -(J^\top J + \mu I)^{-1} J^\top \mathbf{r}$$
 (E2)

Woodbury equivalence:
$$J^\top (J J^\top + \lambda I)^{-1} = (J^\top J + \lambda I)^{-1} J^\top$$
 (E3)

Residual:
$$\mathbf{r}(\boldsymbol{\theta}) = \mathbf{y}_{\text{measured}} - \mathbf{x}(t; \boldsymbol{\theta})$$
 (E4)

Bounded step size:
$$\alpha_t \propto (1 - k_{\text{ibr}}^2)$$
 (E5)

Confidence gate (Gaussian):
$$\gamma = \exp\left(-\frac{\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*\|^2}{2\sigma_\theta^2}\right) \geq \gamma_{\min}$$
 (E6)

Confidence gate (practical):
$$\gamma = 0.35\gamma_R + 0.30\gamma_J + 0.20\gamma_B + 0.15\gamma_P \geq \gamma_{\min}$$
 (E7)

Bounded update (clip):
$$\xi^{(n+1)} = \text{clip}(\xi^{(n)} + \Delta\xi, \xi_{\min}, \xi_{\max})$$
 (E8)

Theorem 2.1:
$$\limsup_n \mathbb{E} \|\hat{\boldsymbol{\theta}}^{(n)} - \boldsymbol{\theta}^*\|^2 = \mathcal{O}(\sigma_{\text{noise}}^2)$$
 (E9)

Condition number (damped):
$$\kappa(J^\top J + \lambda I) = \frac{\sigma_{\max}^2 + \lambda}{\sigma_{\min}^2 + \lambda}$$
 (E10)

Timed Speaker Script (3 Minutes)

What to Say at the TSA Meeting

[0:00–0:30] The Unification Idea

“Chapter 2 is the mathematical spine of the thesis. Before building any individual relay algorithm, I established a unified framework so that every adaptive protection element — 87L, OC, transformer, distance, digital twin — rests on the same mathematical foundation. This slide shows the three-layer structure: an inverse estimator, a confidence gate, and a bounded update law.”

[0:30–1:10] The LM Update Equation

“The core is the Levenberg–Marquardt inverse estimator. The equation on the left is the online update rule: $\hat{\boldsymbol{\theta}}_{t+1}$ equals the current estimate minus a step proportional to the residual r , pre-multiplied by the pseudo-inverse of the Jacobian. The λI damping term is the key: it prevents the step from diverging when the Jacobian is ill-conditioned, which happens routinely at low IBR penetration when the fault current

is weak and parameter identifiability is poor. The bounded step size $\alpha_t \propto (1 - k_{\text{ibr}}^2)$ ensures adaptation freezes at pure IBR operation, preventing parameter drift when the signal is least informative.”

[1:10–1:50] The Confidence Gate

“The Stage-2 gate is essential for protection security. The Gaussian form shown on the slide — $\gamma = \exp(-\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*\|^2/2\sigma_{\hat{\boldsymbol{\theta}}}^2)$ — is the theoretical motivation: a high score means the estimate is close to the true parameters. In the implementation, I approximate this with a four-component score: fit quality γ_R , Jacobian conditioning γ_J , bounds-hit fraction γ_B , and physics plausibility γ_P . Only when the composite $\gamma \geq \gamma_{\text{th}}$ does the algorithm update relay settings. The threshold varies per element: 0.70 for generator OC, 0.85 for line differential.”

[1:50–2:30] Theorem 2.1

“Theorem 2.1 is the formal guarantee. Under two conditions — bounded IBR admittance and strictly positive λ — the update law converges to a ball around the true parameters with radius $\mathcal{O}(\sigma_{\text{noise}})$. This means the algorithm is asymptotically unbiased: the only remaining error is due to measurement noise, not the algorithm itself. Empirically, convergence to $R^2 > 0.999$ is achieved within 15–20 LM iterations, which is approximately 3 ms — well within any protection decision window.”

[2:30–3:00] The Five Methods

“The table on the right shows how five mathematical tools are assigned to the five sub-problems: LM for parameter updates, EKF for continuous state tracking between faults, Wald SPRT for fault detection, Physics-Informed NN for ML classification, and projected gradient for maintaining relay coordination constraints. Chapter 2 establishes all five; subsequent chapters simply specialise them to each protection element by plugging in the physical model for that device.”

Anticipated Q&A — Five Likely Committee Questions

Q1: Why LM and not Adam or other modern optimisers used in deep learning?

Answer: LM is specifically designed for nonlinear *least-squares* problems with a Jacobian structure. Modern ML optimisers (Adam, RMSProp) are designed for stochastic gradient descent on large neural networks with millions of parameters. For the protection problem: (a) the parameter count is 6 (not millions); (b) the model $\mathbf{x}(t; \boldsymbol{\theta})$ is a known physical ODE, not a black-box neural network; (c) the measurement window contains 50–200 samples, not a streaming mini-batch. LM exploits the Jacobian structure to achieve superlinear convergence in 10–20 iterations — Adam would need thousands. Additionally, LM has a rigorous convergence proof (Theorem 2.1); Adam does not have a general convergence guarantee.

Q2: Theorem 2.1 requires bounded IBR admittance — what if Y_{ibr} is time-varying?

Answer: Theorem 2.1 requires $|Y_{ibr}| \leq Y_{\max}$, which is a physical constraint satisfied by any real inverter (the admittance is bounded by the converter filter capacitance and transformer impedance). Time-variation of Y_{ibr} between faults is handled by the EKF in the digital twin, which is a *recursive* extension of LM with a process-noise model for dY_{ibr}/dt . The EKF prediction step provides the equivalent of a time-varying $\lambda(t)$, keeping the estimate within the convergence ball even as Y_{ibr} drifts.

Q3: The slide shows $\gamma = \exp(-\|\hat{\theta} - \theta^*\|^2 / 2\sigma_\theta^2)$ but θ^* is unknown in practice. How is this computed?

Answer: Correct — the exact θ^* is unknown, which is why the theoretical Gaussian form cannot be computed directly. It motivates the approximation: the residual $\|\mathbf{r}(\hat{\theta})\|^2$ proxies for $\|\hat{\theta} - \theta^*\|^2$ via the model fit quality. Specifically, $\gamma_R = 1/(1 + 100(1 - R^2))$ is monotone in R^2 , which in turn is monotone in $\|\mathbf{r}\|^2$. The other three components (γ_J , γ_B , γ_P) provide additional guards that the Gaussian form cannot express, making the practical four-component score *more conservative* than the Gaussian form alone.

Q4: Proposition 2.1 proves settings stay in bounds. Does this mean the relay could become insensitive if settings drift to the wrong end of the allowed range?

Answer: Yes, in principle. The bounded update guarantees safety bounds are never exceeded, but cannot guarantee optimality. This is why the security floor is explicitly included: the lower bound $\xi_{\min}$ is not set to zero but to the minimum sensitivity needed to detect the worst-case bolted fault with a 20% margin. For the generator OC relay, $I_p \geq 1.05I_{\text{load}}^{\max}$ (sensitivity floor) and $I_p \leq 0.80I_{\text{fault},3P}^{\min}$ (coordination floor). The adaptive algorithm operates within this corridor, not at its extremes.

Q5: How does the unified framework handle the diversity of protection elements? Is the parameter vector θ the same for all chapters?

Answer: No — θ is specialised per chapter, which is the whole point of the unified framework: the *structure* is identical (LM + gate + bounded update), but the *content* of θ changes. For 87L: $\theta = \{k_{ibr}, \phi_B\}$ (2 parameters). For SyncOC: $\theta = \{I_{ss}, I_{sub}, \tau_{ac}, \tau_{dc}, \phi_a, t_0\}$ (6 parameters, reduced from 7 by the DC continuity constraint). For the digital twin: $\theta = \{k_{ibr}, Z_{\text{line}}, Y_{ibr}\}$ (3 complex parameters). The chapter's applicability table (Table 2.1) lists all five elements with their specific θ , model, and adapted setting. The unification means one convergence proof (Theorem 2.1) covers all five.

Summary for the TSA Committee

Ch 2 provides one unified mathematical framework (LM inverse estimator + Stage-2 confidence gate + bounded update law) that all SAMBP protection algorithms inherit. Theorem 2.1 proves convergence to a noise-bounded ball around true parameters. The five-method table maps sub-problems (parameter update, state estimation, fault detection, ML, optimisation) to optimal solvers. Every later chapter is an instantiation of this framework with a chapter-specific physical model.